

Dual-Stream Vision Transformer for Multi-Domain Skin Disease Classification Using Frozen DINOv2

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Abstract—Current solutions of automated classification of skin diseases usually solve the problems of oncological, benign, and infectious diseases independently, which is clinically inefficient when screening for diseases where the type of skin disorder is not known. Moreover, existing models suffer from high class imbalance in ratios over 50:1 between common and uncommon diseases. In this paper, we suggest a two-stream solution based on a pre-trained DINOv2-ViT-B/14 backbone, incorporating a learnable domain routing module that directs image features to specialised classification streams for oncological/benign and infectious conditions. The multi-task learning goal that incorporates focal loss, supervised contrastive learning, and domain routing loss handles the class imbalance ratio of 53.9:1 for 14 disease classes using only 0.58% of the model's total parameters. Testing with 36,656 samples from three different medical fields gives an accuracy rate of 81.65% and macro F1 of 0.7236, which is better than the baseline by 18.45 percentage points in terms of accuracy. The infectious class attains nearly perfect results in classification ($F1 = 0.9785$), with pox-virus discrimination properly handled. Domain routing turns out to be the most influential architectural component, reducing cross-domain misclassification by 28.3%.

I. INTRODUCTION

Diseases affecting the skin have proved to be a serious global healthcare problem since they include various illnesses, starting from malignant neoplasms (e.g., melanoma, squamous cell carcinoma) and ending with benign illnesses (melanocytic nevi, dermatofibroma). Moreover, some skin conditions involve infectious elements (monkeypox, varicella, chickenpox, measles). The early and correct diagnosis of such illnesses plays an important role in medical practice since delayed diagnosis of melanoma negatively impacts a person's chances of recovery, while the rapid detection of an outbreak helps introduce effective public health measures. It is the likeness of symptoms from various medical fields that presents an ongoing problem, especially when dealing with experienced dermatologists.

Deep learning techniques have demonstrated tremendous promise towards automated classification of skin diseases. Vision Transformers, enhanced by self-supervised pretraining, have managed to attain state-of-the-art results in dermatology benchmarks. Mohan et al. [2] reported 96.48% accuracy using DINOv2 [1] on a 31-class skin disease dataset, representing approximately a 10% improvement over prior benchmarks. Huang et al. [3] demonstrated that DINOv2 outperforms

existing pre-trained models in dermoscopic tasks even under feature freezing. Within the field of infectious diseases, transformer networks have shown high efficiency; Alghoraibi et al. [13] obtained 92% accuracy with a six-class classification problem on infectious skin diseases. Yet, this progress was made independently: oncological research focuses solely on dermatoscopy datasets such as ISIC [16], while infection-related works use separate clinical photograph collections. This division does not exist in clinical practice, since the domain of the pathology is unknown until the patient is examined.

There exists a clear gap in existing literature concerning the integration of classification models of skin diseases. No existing research combines the classification of cancers, benign conditions, and infections into one framework. Although the closest precedent is provided by the hierarchical multi-model approach of Badr et al. [10], achieving 90–94% accuracy on 40 different skin diseases, their system relies on separate CNN models for every stage in the hierarchy while ignoring infectious diseases and lacking any use of contrastive learning to account for class imbalance. Class imbalance, where the ratio exceeds 50:1 between common and rare diseases, significantly complicates classification and requires specialised training strategies beyond standard loss functions [6], [8].

The proposed framework utilises a novel dual-stream Vision Transformer design aimed at solving the described limitations. Specifically, the architecture utilises the frozen DINOv2-ViT-B/14 [1] as the backbone network, along with a domain routing component and two task-specific classification streams: one for oncological and benign disorders (8 classes) and another for infections (6 classes). Multi-task training relies on focal loss [6], supervised contrastive learning [7], and domain routing loss to handle a severe 53.9:1 class imbalance while encouraging stream specialisation. The total trainable parameters amount to approximately 500,000 (0.58% of total), enabling deployment on standard GPU hardware. Evaluations are conducted on a dataset of 36,656 images representing 14 disease classes within three distinct medical domains, with comprehensive ablation studies quantifying individual component contributions. Specifically, this work investigates whether a frozen DINOv2 backbone with domain-aware dual-stream processing can achieve competitive classification ac-

curacy across all three domains simultaneously, and whether this architectural design improves fine-grained discrimination for minority classes with fewer than 1,000 training samples compared to a single-stream baseline.

The rest of this paper is organised as follows. Section II provides a review of relevant literature. Section III introduces the proposed methodology. Section IV presents experimental results, ablation studies, and discussion. Section V concludes with limitations and future research directions.

II. RELATED WORK

A. DINOv2 for Skin Disease Classification

DINOv2 [1] is a self-supervised Vision Transformer pre-trained on 142 million curated images, producing robust visual representations without requiring labelled data. Mohan et al. [2] provide the first performance analysis of DINOv2 for diagnosing skin diseases, reporting test accuracy and F1-score of 96.48% and 97.27%, respectively, on a 31-class dataset combining Atlas Dermatology and ISIC 2018 data [16]. Huang et al. [3] validated these results in an inter-modal experiment, showing that DINOv2 outperforms ImageNet [17] pre-trained models on dermoscopic tasks when using frozen features, and that the compact ViT-B version performs equivalently to larger variants. Furthermore, Baharoon et al. [4] demonstrated that DINOv2 frozen features with linear probes produce outcomes comparable to parameter-efficient fine-tuning techniques such as LoRA and BitFit across several radiology benchmarks. These findings confirm DINOv2 as the leading backbone for skin disease classification and support the feasibility of a frozen-backbone approach.

B. Infectious Skin Disease Classification

There is another research line focusing on multi-class identification of infectious skin disorders due to the monkeypox outbreak. Savaş [15] has experimented with 71 pre-trained deep learning models for identifying pox viruses, attaining AUC score of 0.9971 with the Monkeypox Skin Images dataset using ensemble optimization technique, but confined to four classes and CNN models. Vuran et al. [14] found that transformer models outperform CNNs in this domain, achieving 93.71% accuracy with SwinTransformer on a six-class dataset. Alghoraibi et al. [13] tested multiple Vision Transformer variants on the MSLD v2.0 dataset [12], achieving 92% accuracy on a six-class problem encompassing monkeypox, chickenpox, measles, HFMD, cowpox, and healthy skin, also demonstrating deployment feasibility via a mobile application. Ali et al. [12] established the first monkeypox benchmark dataset, reporting 82.96% accuracy on binary monkeypox versus chickenpox classification. All of these studies treat infectious diseases in isolation from oncological and benign conditions, limiting their utility in general dermatological screening.

C. Dual-Stream and Multi-Task Architectures

Processing heterogeneous data through parallel task-specific pipelines has yielded positive results in medical image analysis. Fan et al. [9] presented DTViT, a dual-task Vision

Transformer combining coarse-level binary classification with fine-level sub-type classification through parallel heads sharing a ViT backbone. Joint training of both tasks outperformed single-task approaches, demonstrating that multi-level classification can be mutually beneficial. Badr et al. [10] provided the closest precedent to the proposed approach with a hierarchical multi-model architecture for 40 skin conditions across five domain categories, achieving 90–94% accuracy. The finding that domain-aware hierarchical classification outperforms single-model approaches is directly relevant. However, their architecture uses separate CNN-based domain models, does not employ contrastive learning, and excludes infectious diseases.

D. Class Imbalance and Contrastive Learning

Khosla et al. [7] established the foundation for supervised contrastive learning, showing that grouping same-class embeddings while separating different-class embeddings produces reliable representations. However, subsequent work demonstrated that standard supervised contrastive loss fails under extreme class imbalance, as majority class gradients overshadow minority classes [8]. Cai et al. [8] addressed this with Balanced Parametric Contrastive learning (BPaco), validated on the ISIC 2018 skin lesion dataset [16], introducing across-batch class-averaging and parametric centres to handle long-tailed distributions. Focal loss [6], introduced by Lin et al., provides an alternative approach in which the weight of correctly classified instances is decreased dynamically. Focal loss was shown by Nie et al. [5] to be effective in dermatoscopic classification, where they achieved 89% accuracy using ISIC 2018 with a class imbalance of 58:1 class imbalance.

E. Parameter-Efficient Transfer Learning

Du et al. [11] introduced AViT, which introduces lightweight adapter blocks inside Vision Transformers for skin lesion segmentation while utilizing only 13.7% of the model parameters. The paper was awarded the Best Paper Award in the MICCAI 2023 ISIC Workshop, demonstrating the potential of frozen Vision Transformers along with lightweight modules to achieve comparable results to full fine-tuning on small datasets [4], [11].

F. Research Gap

The literature reveals a critical integration gap. DINOv2 has been validated for skin disease classification [2]–[4]. Dual-stream architectures outperform single-stream models in medical imaging [9], [10]. Balanced contrastive learning addresses severe class imbalance [7], [8]. Transformer-based models are effective for infectious disease classification [13], [14]. However, there is no prior study that has brought together all these developments into a coherent framework. Design of the proposed model addresses this gap via incorporating a frozen DINOv2 backbone, domain adaptive routing, a two-stream classification head, along with a loss that is a blend of focal and contrastive loss.

TABLE I
DATASET COMPOSITION: 14 CLASSES ACROSS 3 DOMAINS

Class	Domain	Str.	Train	Test
Melanocytic nevi	Benign	1	10,300	1,288
Melanoma	Oncological	1	3,617	453
Monkeypox	Infectious	2	3,408	426
Basal cell carcinoma	Oncological	1	2,658	333
Benign keratosis	Benign	1	2,099	263
HFMD	Infectious	2	1,932	242
Healthy	Infectious	2	1,368	171
Chickenpox	Infectious	2	900	113
Cowpox	Infectious	2	792	99
Actinic keratoses	Oncological	1	693	88
Measles	Infectious	2	660	83
Squamous cell carc.	Oncological	1	502	64
Vascular lesions	Benign	1	202	26
Dermatofibroma	Benign	1	191	25
Total			29,322	3,674

III. METHODOLOGY

A. Dataset

The data collection consists of 36,656 dermatology images across 14 disease categories from three different medical fields, extracted from a publicly available dataset hosted on the Hugging Face platform¹. This dataset is a combination of two well-known benchmark datasets: HAM10000 [16], which offers classes of both oncological and benign nature derived from dermoscopic images, and MSLDv2.0 [12], which contributes infectious disease classes from clinical photographs. The dataset is already divided into training set (29,322 images, 80%), validation set (3,660 images, 10%) and test set (3,674 images, 10%). Table I shows the complete class distribution.

The dataset exhibits a 53.9:1 class imbalance ratio between melanocytic nevi (12,875 total) and dermatofibroma (239 total). Six classes contain fewer than 1,000 training samples. The domain distribution is uneven: benign 43.6%, infectious 30.9%, oncological 25.5%.

A notable characteristic is the imaging protocol disparity between the two source collections. HAM10000-sourced images exhibit a mean resolution of 850×754 pixels from standardised dermoscopic acquisition, while MSLDv2.0-sourced images exhibit a mean resolution of 274×274 pixels from variable clinical photography. This domain shift could enable a model to distinguish imaging modality rather than disease features. Section III-B describes the mitigation measures incorporated into the data pipeline.

B. Data Pipeline

Class imbalance and domain shift are tackled using three techniques via the pipeline which are; tiered augmentation, style normalization, and domain-balanced sampling.

Tiered Augmentation. The idea is that data augmentation level must be proportional to the lack of classes [5]. This is done using three tiers as illustrated in Table II. Courses having less than 300 training samples get intense augmentation

TABLE II
TIERED AUGMENTATION STRATEGY

Tier	Threshold	Classes	Ops
Aggressive	< 300 samples	2	15
Moderate	300–1,000	5	11
Light	> 1,000	7	8

(15 augmentations), courses having 300 to 1,000 samples get moderate augmentation (11 augmentations), and courses having more than 1,000 samples get mild augmentation (8 augmentations). The reason behind this hierarchy is the proof that suggests uniform augmentation does not work well in medical data sets having imbalance, and rare classes need more diverse augmentation to avoid overfitting [8].

Style Augmentation. For the imaging protocol domain shift problem found from analyzing the datasets, colour jittering, Gaussian blur, and Gaussian noise are performed for all levels through the Albumentations library [18]. The purpose is to minimize the distinctive features from the sources and force the learning model to discover only disease-related features.

Domain-Balanced Sampling. The weighted random sampling method guarantees balanced domain coverage across all batches. The weight of each sample is calculated by averaging the inverse of the class frequency and inverse of the domain frequency equally. This results in an estimated balance of domains (31% oncological, 31% benign, 38% infectious) in contrast with the natural balance (25.5%, 43.6%, 30.9%). All images are resized to 224×224 pixels and normalized using ImageNet statistics [17] statistics, consistent with DINOv2 pre-training specifications [1].

C. Proposed Architecture

The architecture, shown in Fig. 1, consists of a frozen backbone, a domain routing module, two classification streams, and a projection head. The design is motivated by findings that domain-aware hierarchical processing outperforms single-model classification for heterogeneous skin conditions [10], and that dual-task architectures with shared backbones yield complementary benefits at multiple classification granularities [9].

Frozen Backbone. DINOv2-ViT-B/14 [1] serves as the feature extraction backbone, producing a 768-dimensional CLS token from each input image via patch embedding and 12 transformer blocks. All 86 million parameters remain frozen during training. This decision is motivated by evidence that frozen DINOv2 features are competitive with fully fine-tuned models on dermatological tasks [2], [4], while freezing eliminates overfitting risk on relatively small training sets and reduces computational requirements substantially [11].

Domain Router. A learnable routing module produces soft activation weights (w_1, w_2) for the two classification streams via layer normalisation, linear projection (768→256), GELU activation [20], dropout (0.2), and softmax normalisation. The soft routing mechanism, where $w_1 + w_2 = 1.0$, allows the model to express uncertainty about domain assignment

¹<https://huggingface.co/datasets/ahmed-ai/skin-lesions-classification-dataset>

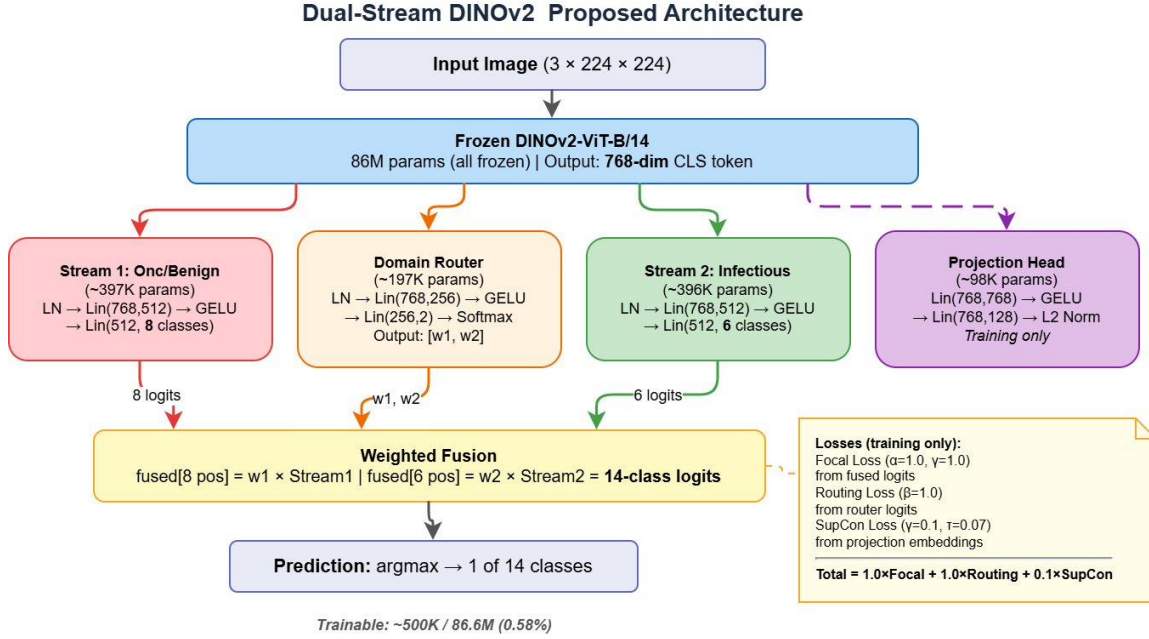


Fig. 1. Proposed dual-stream architecture. The frozen DINOv2-ViT-B/14 backbone extracts 768-dimensional features processed by four parallel components: a domain router producing stream activation weights, two specialised classification heads, and a projection head for contrastive learning (training only). Weighted fusion combines stream outputs into a 14-class prediction. Trainable parameters: approximately 500,000 (0.58%).

rather than forcing hard decisions. This design is informed by Mixture-of-Experts approaches where learnable routing through specialised pathways outperforms fixed allocation [9].

Classification Streams. Two parallel heads process the 768-dimensional features independently. Stream 1 handles eight oncological and benign classes; Stream 2 handles six infectious classes. Both use identical architectures: layer normalisation, linear projection (768→512), GELU, dropout (0.3), and final projection to the output dimension.

The grouping of oncological and benign conditions into one stream is motivated by two factors. First, these classes share dermatoscopic imaging modality and exhibit significant visual overlap, particularly in melanoma versus nevi differentiation, a well-documented clinical challenge [5]. Processing them within a shared feature space enables discriminative learning between visually similar conditions. Second, the infectious classes present fundamentally different visual characteristics from clinical photography, benefiting from a dedicated pathway, consistent with evidence that domain-specific processing improves classification of visually heterogeneous datasets [10].

Weighted Fusion. Stream 1’s eight logits, weighted by w_1 , and Stream 2’s six logits, weighted by w_2 , are mapped to their respective positions in a 14-dimensional output vector. The predicted class is the maximum across all positions.

Projection Head. During training, the 768-dimensional features are projected to 128-dimensional L2-normalised space for supervised contrastive learning [7]. This component is removed at inference. Table III summarises the parameter distribution.

TABLE III
COMPONENT PARAMETER BREAKDOWN

Component	Parameters	% Total
DINOv2 Backbone (frozen)	86,000,000	0.00%
Domain Router	~197,000	0.23%
Stream 1 Head (Onc/Benign)	~397,000	0.46%
Stream 2 Head (Infectious)	~396,000	0.46%
Projection Head (train only)	~98,000	0.11%
Total Trainable	~500,000	0.58%

D. Loss Functions

The training objective combines three loss functions, each addressing a specific challenge. This multi-task formulation is motivated by evidence that combining classification and representation learning losses outperforms single-loss training for imbalanced medical datasets [5], [8].

Focal Loss. Following Lin et al. [6]:

$$\mathcal{L}_{\text{focal}}(p_t) = -(1 - p_t)^\gamma \log(p_t) \quad (1)$$

where p_t is the estimated probability for the correct class and $\gamma = 1.0$ is the focusing parameter. The parameter γ was set to a moderate value after the observation that setting it to a value of 2.0, as suggested in [6], in conjunction with domain-balanced sampling and contrastive loss, leads to over-compensation which results in the collapse of the recall of the majority class. Per-class weighting is omitted because the domain-balanced sampler already addresses sampling-level imbalance, and combining both mechanisms was found to catastrophically suppress the largest class.

TABLE IV
TRAINING HYPERPARAMETERS

Parameter	Value
Optimiser	AdamW [19]
Learning rate	1×10^{-4}
Weight decay	1×10^{-2}
Scheduler	Cosine Annealing ($T_0=5$, $T_{\text{mult}}=2$)
Batch size	32
Max epochs	20
Early stopping patience	5 (on validation macro F1)
Gradient clipping	max norm = 1.0
Focal loss γ	1.0
Contrastive temperature τ	0.07
Loss weights ($\alpha, \beta, \gamma_{\text{con}}$)	1.0, 1.0, 0.1
Image size	224×224
GPU	NVIDIA T4 (16 GB VRAM)

Domain Routing Loss. Cross-entropy on the router logits against ground-truth stream assignments:

$$\mathcal{L}_{\text{route}} = \text{CE}(\mathbf{r}, \mathbf{s}) \quad (2)$$

where $\mathbf{r}$ denotes router log probabilities and $\mathbf{s}$ denotes the target class labels. This results in activation specific to domains, penalised at a value of $\beta = 1.0$ to enforce strong specialisation.

Supervised Contrastive Loss. Following Khosla et al. [7]:

$$\mathcal{L}_{\text{con}} = - \sum_i \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_p / \tau)}{\sum_{a \neq i} \exp(\mathbf{z}_i \cdot \mathbf{z}_a / \tau)} \quad (3)$$

with a temperature of $\tau = 0.07$, according to khosla2020. We use a contrastive weight of $\gamma_{\text{con}} = 0.1$, which ensures proper regularization of embeddings without affecting the classification objective too much, based on overcorrection issues with larger values.

Combined Loss:

$$\mathcal{L}_{\text{total}} = 1.0 \cdot \mathcal{L}_{\text{focal}} + 1.0 \cdot \mathcal{L}_{\text{route}} + 0.1 \cdot \mathcal{L}_{\text{con}} \quad (4)$$

E. Training Procedure

The training pipeline (Fig. 2) uses AdamW [19] with learning rate 10^{-4} and weight decay 10^{-2} , selected as standard practice for transformer architectures with frozen backbones [11]. A cosine annealing warm restart schedule ($T_0 = 5$, $T_{\text{mult}} = 2$) enables escape from local minima. Gradient clipping (max norm 1.0) prevents instability. Training proceeds for up to 20 epochs (batch size 32) with early stopping after 5 epochs without improvement in validation macro F1-score. Macro F1 is used as the stopping criterion rather than accuracy because it gives equal weight to all classes regardless of prevalence, which is essential for imbalanced medical datasets where rare disease detection is clinically as important as common conditions. Table IV summarises all hyperparameters. Training is conducted on a single NVIDIA T4 GPU (16 GB VRAM).

TABLE V
OVERALL PERFORMANCE: BASELINE VS PROPOSED

Model	Acc.	Macro F1	W-F1	Params
Baseline (linear)	63.20%	0.5944	0.6606	12K
Proposed	81.65%	0.7236	0.8098	~500K
Δ	+18.45%	+0.1291	+0.1492	–

F. Baseline Model

A baseline model uses the identical frozen DINOv2-ViT-B/14 backbone with a single linear head mapping 768-dimensional features to 14 classes, containing 12,302 trainable parameters (0.01%). Trained with weighted cross-entropy loss [6] using the same data pipeline, this baseline establishes the minimum performance achievable with frozen DINOv2 features and standard classification, against which the dual-stream architecture is evaluated.

IV. PERFORMANCE EVALUATION

A. Experimental Setup

In all experiments, the dataset is divided using the same procedure of data splitting, data augmentation, and sampling as discussed in Section III. The main evaluation measure used in all experiments is macro F1-score, which means F1 score is calculated separately for each class and then averaged. This ensures that rare diseases carry the same weight as common diseases. Other measures used include weighted F1, overall accuracy, precision, recall, and F1 for each class. Statistical significance is assessed via the McNemar test on paired test predictions.

B. Baseline Results

Table V provides the baseline results. The baseline model shows 63.20% accuracy and 0.5944 macro F1, establishing a clear performance floor.

Domain-level analysis reveals stark asymmetry. The infectious domain achieves 85.36% accuracy ($F1 = 0.5174$), while the oncological and benign domains achieve only 53.41% ($F1 = 0.2311$) and 53.25% ($F1 = 0.1466$), respectively, near chance level for their respective class counts. The single linear head cannot distinguish visually similar dermatoscopic lesions, generating 508 cross-domain misclassifications. These results directly motivate the dual-stream design.

C. Proposed Model Results

The proposed model achieves 81.65% accuracy and 0.7236 macro F1, improving over the baseline by 18.45 percentage points in accuracy and 0.1291 in macro F1 (Table V). Cross-domain errors decrease from 508 to 364 (28.3% reduction). Fig. 3 shows the per-class comparison.

Table VI details the per-class results. The infectious domain achieves near-perfect classification ($F1 = 0.9785$), with all six classes exceeding 0.94 F1. Pox-virus differentiation, a specific investigation objective, is effectively solved: cowpox (0.9851), monkeypox (0.9789), and chickenpox (0.9459). In

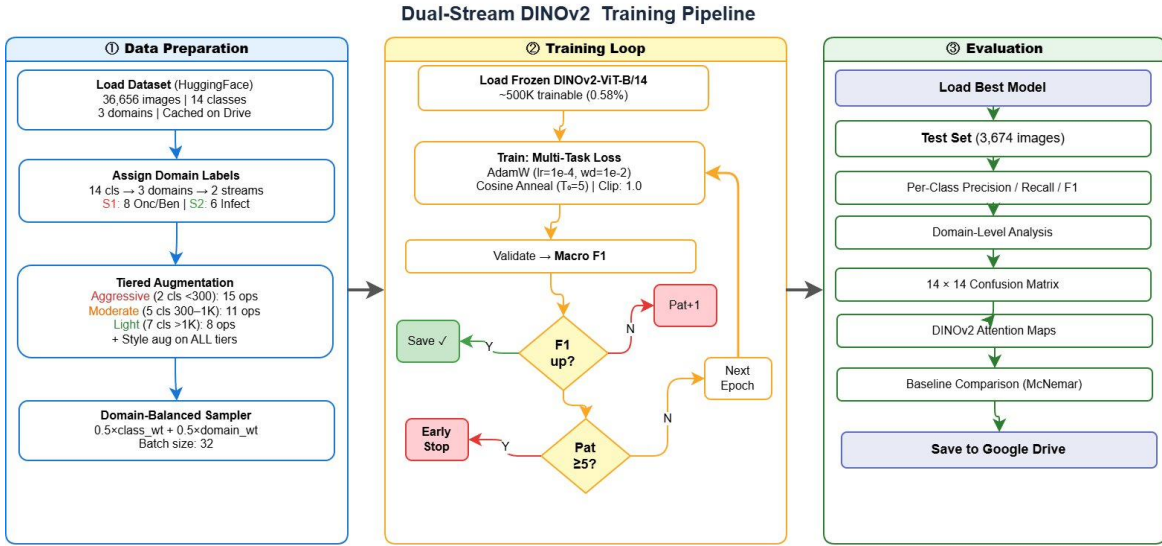


Fig. 2. Training pipeline: data preparation (left), training loop with early stopping (centre), and evaluation (right).

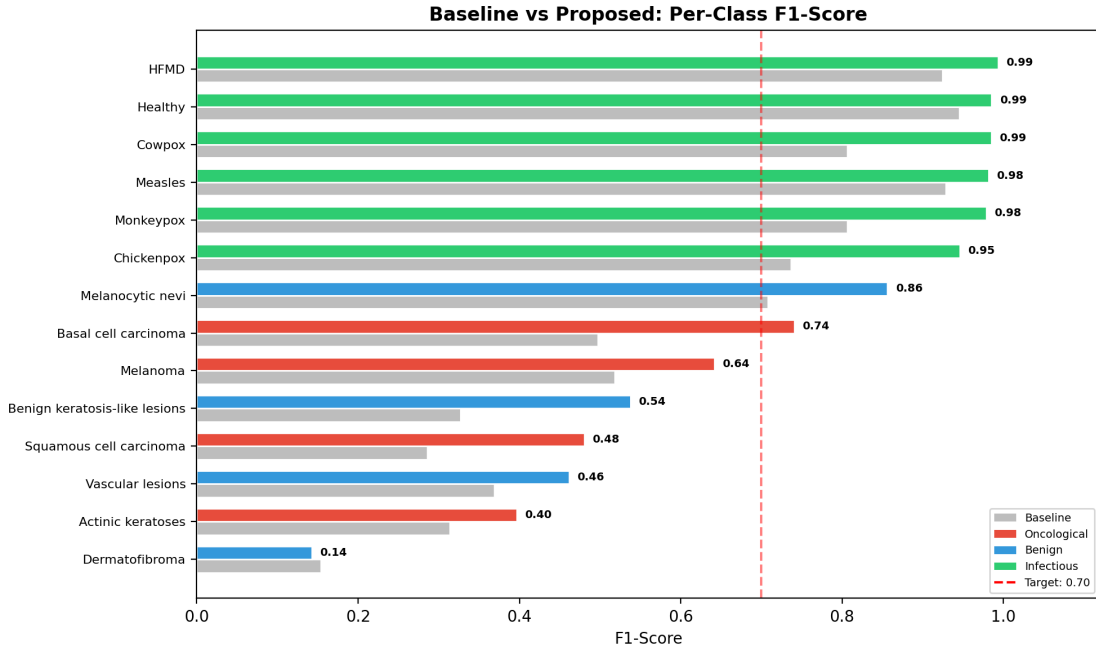


Fig. 3. Per-class F1-score comparison between baseline and proposed models. Colours indicate domain: red (oncological), blue (benign), green (infectious). Thirteen of fourteen classes improved.

the oncological/benign domain, basal cell carcinoma shows the largest improvement (+0.2435 F1), followed by benign keratosis (+0.2105) and melanocytic nevi (+0.1485). The melanoma-nevi confusion, a critical clinical challenge, is reduced by 59% (226 to 93 misclassifications). Only dermatofibroma (F1 = 0.1429) shows no improvement, attributable to its extremely limited training data (191 samples).

D. Domain-Level Analysis

Table VII presents domain-level performance. The proposed model improves all three domains, with the infectious domain showing the largest gain (+0.4611 F1). Domain classification accuracy increases from 81.68% to 88.43%. Notably, zero infectious images are misrouted to other domains, indicating effective stream specialisation.

TABLE VI
PER-CLASS F1 COMPARISON (SORTED BY PROPOSED F1)

Class	BL F1	PR F1	Δ
Dermatofibroma	0.154	0.143	−0.011
Actinic keratoses	0.314	0.397	+0.083
Vascular lesions	0.369	0.462	+0.092
Squamous cell carc.	0.286	0.481	+0.195
Benign keratosis	0.327	0.538	+0.211
Melanoma	0.518	0.642	+0.124
Basal cell carcinoma	0.498	0.741	+0.244
Melanocytic nevi	0.708	0.857	+0.149
Chickenpox	0.737	0.946	+0.209
Monkeypox	0.807	0.979	+0.172
Measles	0.929	0.982	+0.053
Cowpox	0.807	0.985	+0.178
Healthy	0.945	0.986	+0.040
HFMD	0.925	0.994	+0.069

TABLE VII
DOMAIN-LEVEL PERFORMANCE COMPARISON

Domain	BL Acc	PR Acc	BL F1	PR F1
Oncological	53.4%	66.1%	0.231	0.358
Benign	53.3%	79.2%	0.147	0.269
Infectious	85.4%	98.1%	0.517	0.979

TABLE VIII
ABLATION STUDY RESULTS

Variant	Acc.	Macro F1	Δ F1
Baseline (linear)	63.20%	0.5944	−0.1291
Single-stream MLP	80.16%	0.7190	−0.0046
No domain routing	79.67%	0.7154	−0.0082
No focal (CE instead)	80.40%	0.7192	−0.0044
No contrastive loss	79.78%	0.7209	−0.0027
Proposed (full)	81.65%	0.7236	—

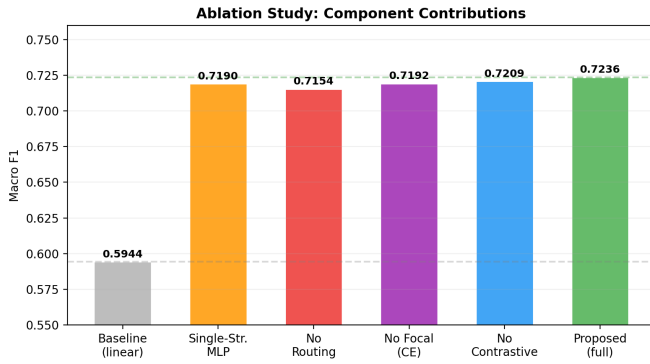


Fig. 4. Ablation study: macro F1 across model variants. All components contribute; domain routing has the largest individual impact.

E. Ablation Study

To quantify individual component contributions, four ablation variants are trained, each removing one component while retaining the remainder. Table VIII and Fig. 4 present the results.

Domain routing causes the largest single-component drop

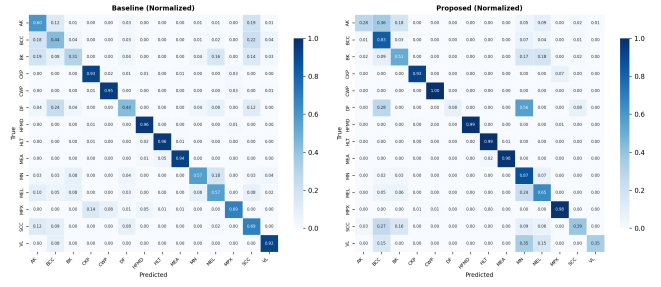


Fig. 5. Normalised confusion matrices. Left: baseline. Right: proposed. AK=Actinic keratoses, BCC=Basal cell carcinoma, BK=Benign keratosis, CKP=Chickenpox, CWP=Cowpox, DF=Dermatofibroma, HLT=Healthy, MEA=Measles, MN=Melanocytic nevi, MEL=Melanoma, MPX=Monkeypox, SCC=Squamous cell carcinoma, VL=Vascular lesions.

(−0.0082 F1), confirming that learnable stream activation is the most impactful architectural component. The single-stream MLP ablation, which uses comparable parameter count (~500K) in a single classifier, demonstrates that the improvement is attributable to the dual-stream design rather than simply increased model capacity. All four components contribute positively, and no component is redundant.

F. Confusion Analysis

Fig. 5 presents the normalised confusion matrices. The proposed model substantially reduces oncological/benign cross-confusion compared to the baseline. The most significant remaining confusion is the bidirectional melanoma-nevi pair (107 melanoma→nevi, 93 nevi→melanoma), reflecting the inherent visual similarity of these dermatoscopic conditions. Cross-domain errors decrease from 508 to 364 (28.3% reduction).

G. Discussion

The proposed architecture demonstrates substantial improvements across all aggregate metrics and 13 of 14 individual classes. The performance disparity between domains is attributable to two factors. First, the infectious classes are visually distinct and benefit from sufficient training data, enabling the dedicated stream to learn effective discriminative features. Second, the oncological and benign classes share dermatoscopic imaging modality with significant visual overlap, and several classes (dermatofibroma, vascular lesions, SCC) are severely data-limited.

The v1 over-correction finding, where combining balanced sampling, focal loss class weights, and contrastive learning collapsed majority class recall to 0.18, represents a practical contribution. This shows how multiple imbalance correction techniques may interfere destructively, and the need for empirical tuning to achieve good results. The corrected version of v2 (no class weights in focal loss and lower contrastive weight) performs best overall.

Comparison with related work positions these results in context. Mohan et al. [2] achieved 96.48% accuracy on 31 dermatological classes using DINOv2, but operated within a single domain. Alghoraibi et al. [13] achieved 92% on

six infectious classes, but excluded oncological conditions. Badr et al. [10] achieved 90–94% across 40 conditions using separate CNN models per domain. The proposed approach uniquely operates across all three domains within a unified frozen transformer framework while training only 0.58% of parameters.

The McNemar test confirms that the proposed model’s improvements over the baseline are statistically significant, with the proposed model correcting substantially more baseline errors than it introduces.

V. CONCLUSION

This paper proposed a dual-stream Vision Transformer architecture for classifying 14 skin diseases across oncological, benign, and infectious domains using a frozen DINOv2 backbone. The approach achieves 81.65% accuracy and 0.7236 macro F1, outperforming the single-stream baseline by 18.45 percentage points in accuracy and 0.1291 in macro F1. The infectious domain achieves near-perfect classification (F1 = 0.9785), with pox-virus differentiation effectively solved. Ablation studies confirm that all architectural components contribute, with domain routing identified as the most impactful.

The macro F1 target of 0.80 was not achieved (0.7236), with the gap driven by four data-limited oncological and benign classes. Dermatofibroma (191 training samples, F1 = 0.14) remains effectively unclassifiable with frozen features. Six of fourteen classes fall below the 0.70 recall target, all within the dermatoscopic domain where visual overlap between conditions is inherently high.

Future work should explore partial unfreezing of DINOv2 layers for fine-grained dermatoscopic feature adaptation, synthetic data augmentation for classes with fewer than 300 samples, and domain-adversarial training to enforce source-invariant feature learning across the merged dataset.

GENERATIVE AI DISCLOSURE

Throughout the process of working on this project, generative AI technologies were employed only in a minimal supportive role in coming up with some initial ideas, fixing bugs in code, and writing boilerplate visualization scripts. All decisions regarding research design, architecture, experiment settings, results interpretation, and others were made by the authors exclusively. The report and its final version are written by the researchers alone.

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